Joe Kanjirathingal

Data Science and Machine Learning
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SUMMARY

B.Tech Artificial Intelligence student graduating in March 2027, seeking data analyst, data science and machine learning internships. Analysed 26.5 million wholesale market records to trace Kerala vegetable price shocks and built a spike-warning model; project experience in Python, SQL, Power BI, Excel, web scraping, statistics, NLP classification and forecasting, with reproducible evaluation.

EDUCATION

B.Tech in Artificial Intelligence | Sep 2023 - Mar 2027 (expected)
Vidya Academy of Science and Technology, Thrissur, Kerala

TECHNICAL SKILLS

Programming and Data: Python, SQL (window functions, CTEs), DuckDB, Parquet, pandas, NumPy, data validation
Analytics and BI: Power BI (DAX, Power Query), Excel (PivotTables, XLOOKUP, SUMIFS), time series, A/B tests, forecasting
Machine Learning: scikit-learn, PyTorch, Hugging Face Transformers, ONNX, NLP, classification, calibration
Visualization and Applications: Flourish, Chart.js, Three.js, Leaflet, web scraping (Scrapling), FastAPI, React
Databases and Tools: PostgreSQL, SQLite, Supabase, Git, GitHub, Vercel

SELECTED PROJECTS

Kerala Vegetable Prices - Price Shocks, Market Gaps and Spike Warning | Live site
Nov 2026 | Python, SQL, DuckDB, scikit-learn, statsmodels, Power BI, Excel, Scrapling
- Built a DuckDB pipeline over 26.5M Agmarknet records (5 GB) with 13 data tests; merged 3,860 market codes into 2,007.
- Found Kerala paid 1.9-3.3x the all-India wholesale price and 20%+ gaps between markets 40 km apart; built a Power BI report (DAX star schema) and an Excel workbook with PivotTables.
- Trained a spike-warning model on a chronological split: 0.33 PR-AUC versus 0.04 chance, warning of 4 of 5 held-out spikes a median 16 days early; scraped news features hurt and were dropped.

Retail Analytics Lab - E-commerce Data Warehouse and Analysis | Live site
Oct 2026 | Python, SQL, DuckDB, Parquet, NumPy, SciPy, Excel
- Built a reproducible Python and DuckDB pipeline and star schema over 110M e-commerce events (4.6 GB); 14 data tests reconcile rows and $505.1M of revenue from raw files to every output.
- Detected a 48-hour purchase-tracking outage that created a false 53% weekend revenue effect; added automated gap detection and excluded the affected days from conversion analysis.
- Wrote SQL marts (window functions, cohorts, Pareto) and a decision memo: 42% of buyers drive 79% of revenue; 13-day forecast at 5.6% MAPE versus a 5.9% seasonal-naive baseline.

QuoteCheck - Supplier Quotation Review and Product Matching | Project
Oct 2026 - Present | Python, PyTorch, FastAPI, React, PostgreSQL, ONNX, Vercel
- Built a quotation-review app with PDF/XLSX extraction, source-linked corrections, exact pricing and exports.
- Fine-tuned MiniLM on 19,835 product pairs; achieved 74.3% validation average precision and 64.7% F1 on a 4,500-pair WDC Products benchmark, versus 43.4% F1 for the logistic baseline.
- Deployed ONNX CPU inference on Vercel with PostgreSQL; verified by 247 automated tests and 3 browser workflows.

IntentDesk - NLP Classification and Evaluation | GitHub
Oct 2026 | Python, scikit-learn, PyTorch, FastAPI, SQLite
- Built an NLP workflow for 150 intent categories using frozen pretrained MPNet embeddings, a downstream classifier and probability calibration.
- 96.24% accuracy on 4,500 CLINC150 test requests versus 91.96% for a lexical baseline; kept human review after unsupported-request detection failed.

Kerala Crime Intelligence Map - Data Dashboard | Live demo
Sep 2026 - Present | Python, JavaScript, Leaflet, Chart.js
- Built an interactive dashboard of published Kerala Police statistics with district comparisons, trend charts and population-adjusted rates; reconciled district totals against the state total.

TRAINING

Python Full-Stack Development | SMEClabs, Kochi | Jun 2026 - Dec 2026 (expected)
Six-month programme covering Python, SQL, Django, HTML/CSS and JavaScript.